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EV Pulse NC: A Data-Driven Framework for Equitable and Demand-Driven EV Charging Infrastructure Investment Prioritization in North Carolina

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\$109 Million. 100 Counties. No Data-Driven County Ranking.

\$109M

NC's federal National Electric Vehicle Infrastructure (NEVI) Formula Program funding

No county ranking

The NC Department of Transportation (NCDOT) has a federally approved NEVI plan, but no publicly available, data-driven method for ranking counties

Feb 2026

NCDOT revises its NEVI plan, narrowing the corridor buildout and shifting funds toward rural and community charging

Efficiency vs. equity

Demand-driven allocation reinforces existing concentration; equity-driven allocation can place stations where use is too low to sustain them.

Five Questions Drive the Work

1	Forecast Accuracy	How accurate are SAS Model Studio's county-level battery electric vehicle (BEV) forecasts against unseen data?
2	Demand-Supply Gaps	Where are the largest demand-supply gaps?
3	Sub-County Inequality	How do ZIP-level density patterns reveal priorities county aggregation obscures?
4	Commuter Flows	How do commuting flows reshape demand once workplace needs are layered onto residential data?
5	Defensible Rankings	Can a weighted scoring equation translate these layers into defensible recommendations?

The BEV Fleet Grew 1,727% Since 2018. Ports Have Not Kept Pace.

5,165 → 94,371

BEVs, Sep 2018 → Jun 2025

53.8%

Compound annual growth

Gini 0.805

County-level BEV ownership is highly concentrated; Wake alone holds more BEVs than the bottom 60 counties combined

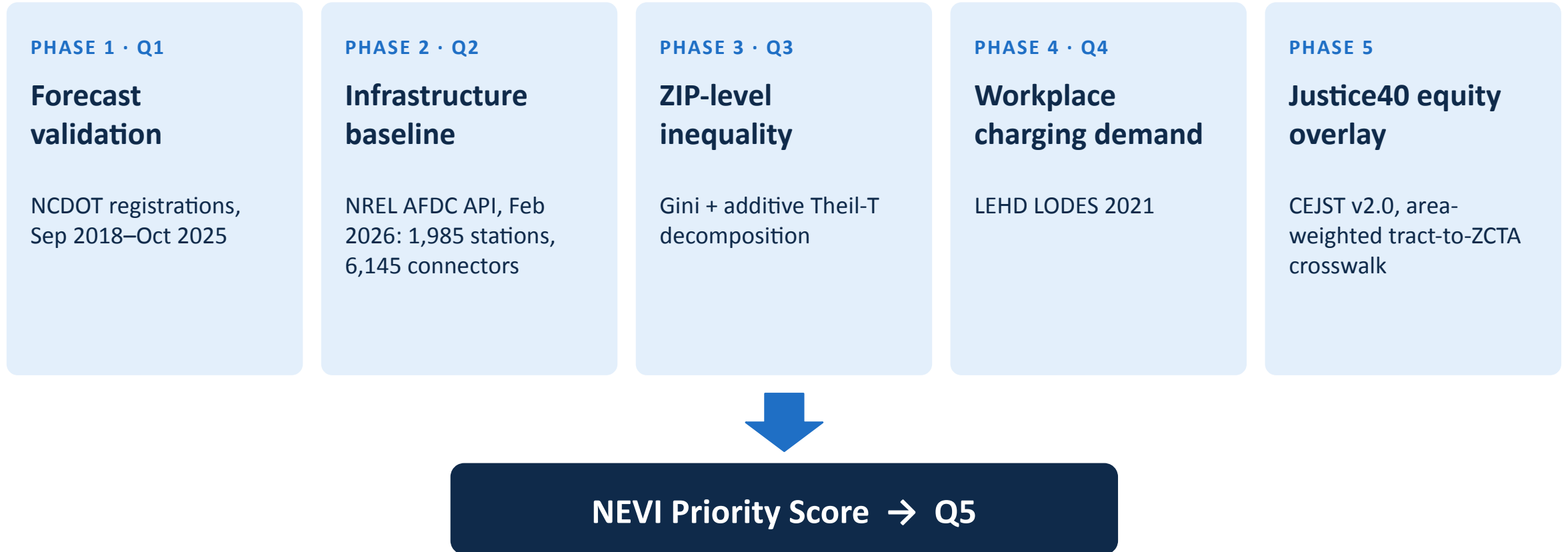
16.9

BEVs per public port (Feb 2026) vs. International Energy Agency (IEA) global benchmark ≈10

Top 10 counties hold **73%** of the statewide BEV fleet

Data as of Feb 2026: NCDOT registrations through Oct 2025; Alternative Fuels Data Center (AFDC) stations as of Feb 2026

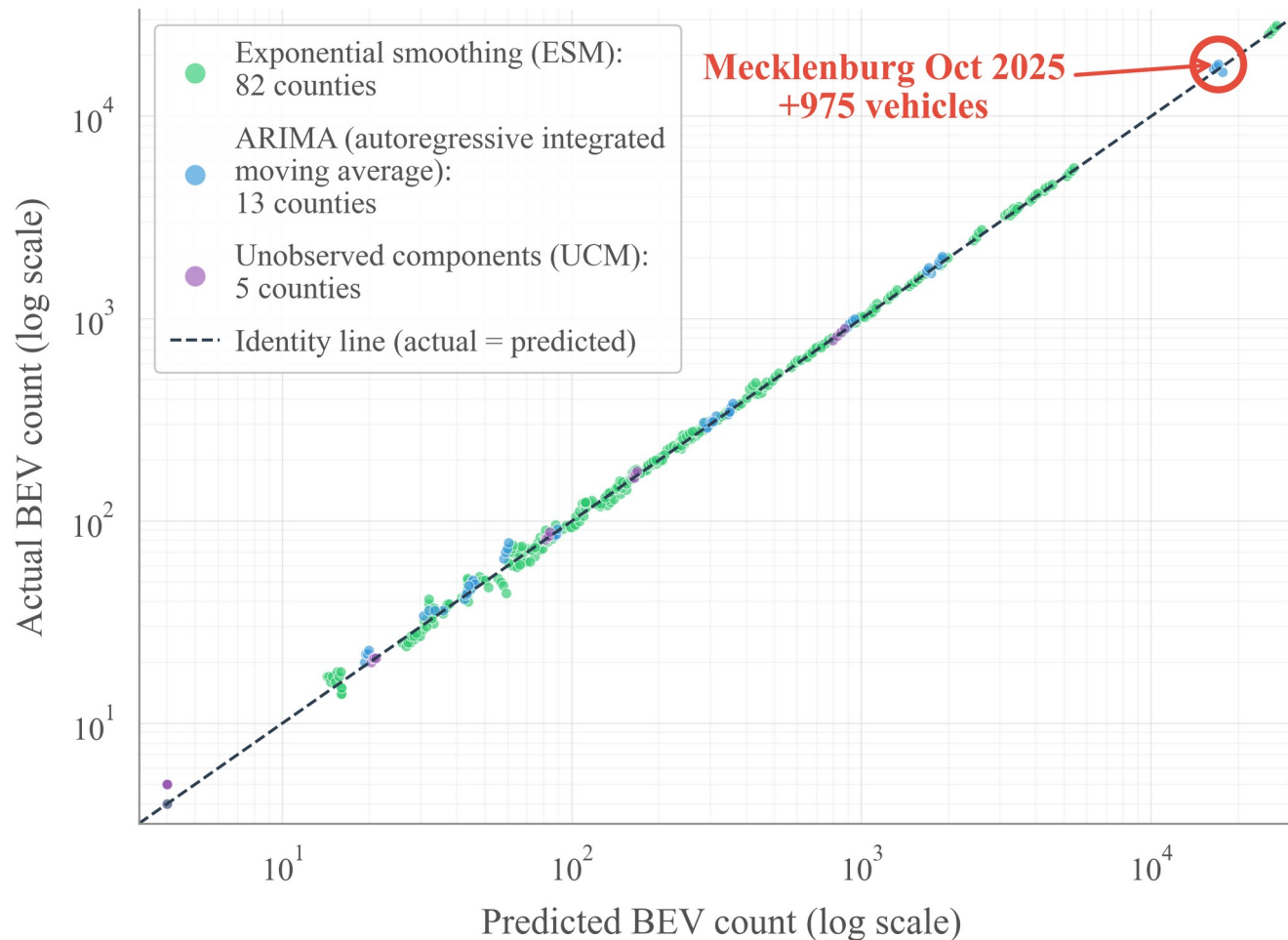
Five Analytical Phases, One Scoring Framework



All public data · 8,600 county-month observations · SAS Model Studio forecasts (fall 2025), validated in Python (spring 2026)

NREL: National Renewable Energy Laboratory · AFDC: Alternative Fuels Data Center · API: application programming interface · LEHD LODES: Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics · CEJST: Climate and Economic Justice Screening Tool · ZCTA: ZIP Code Tabulation Area

Forecasts Hold Out of Sample, and Adoption Accelerated After 2022



Mean absolute percentage error (MAPE) 4.34%
on a true holdout (n = 400 county-months, Jul–Oct 2025, all 100 counties)

69.00% of actuals exceeded forecast (mean bias +18.22 vehicles per county-month)

Chow test: $F = 1,268.35$, $p < 1 \times 10^{-6}$ at the Aug 2022 Inflation Reduction Act passage

95% interval coverage **75.50% raw → 93.75%**
after county bias correction: a centering problem, not a width problem

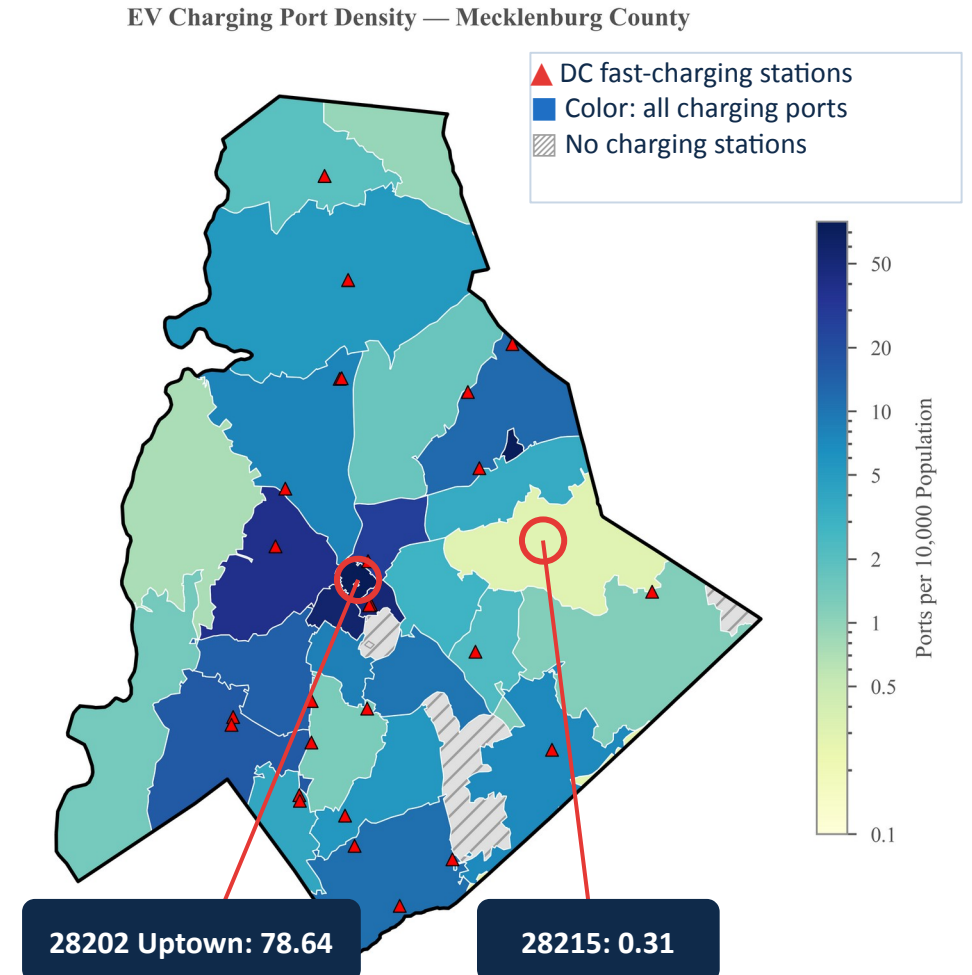
A 250-Fold Charging Gap Inside One County

58.5% of NC ZIP areas (499 of 853) have zero stations: **2.2 million people** (21.1% of the state)

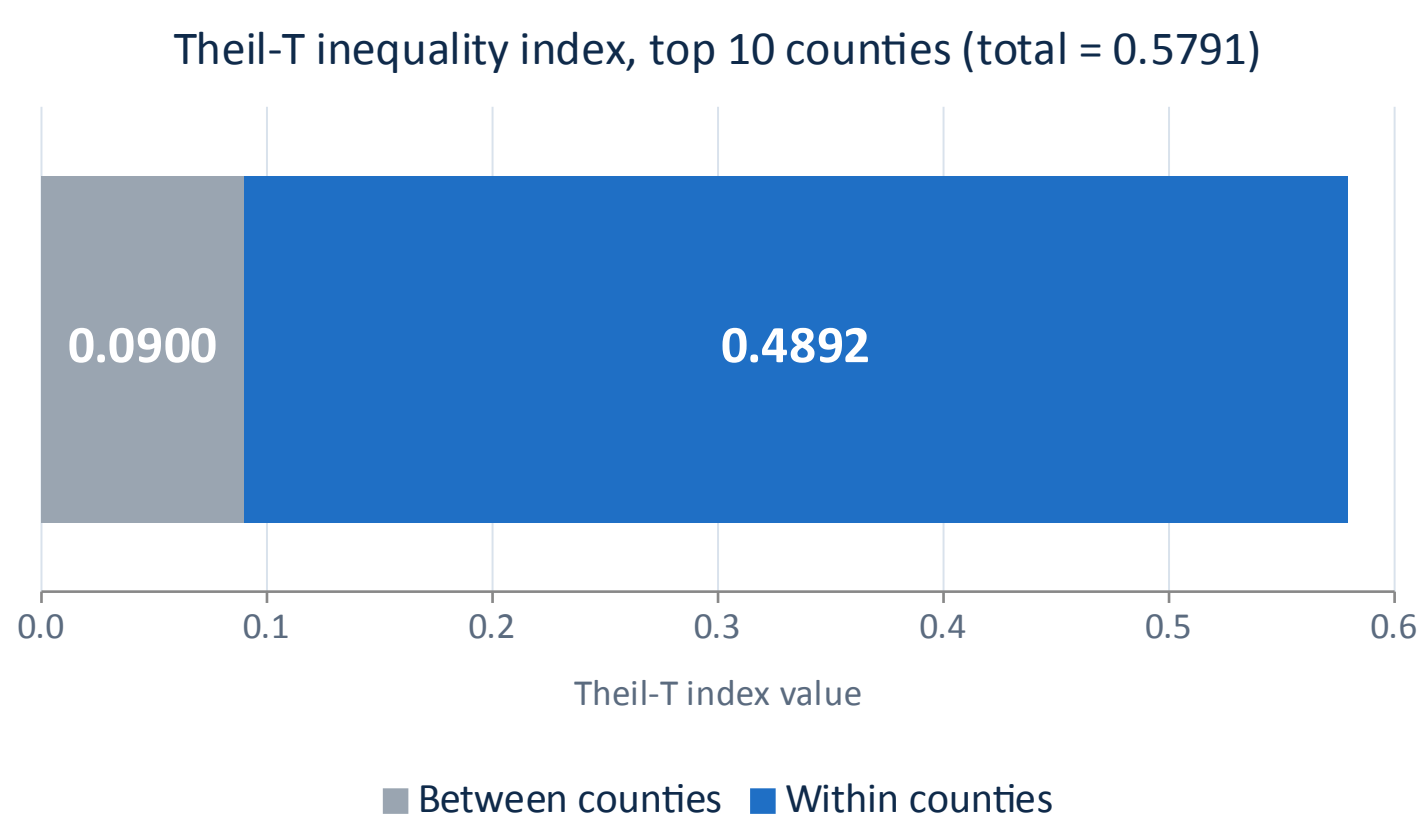
Charlotte 28202: **78.64** ports per 10,000 residents

Charlotte 28215, 14 miles east: **0.31** (64,713 residents, 2 ports)

The gap is not rural alone



84.5% of Charging Inequality Is Within Counties



84.5%

within

15.5%

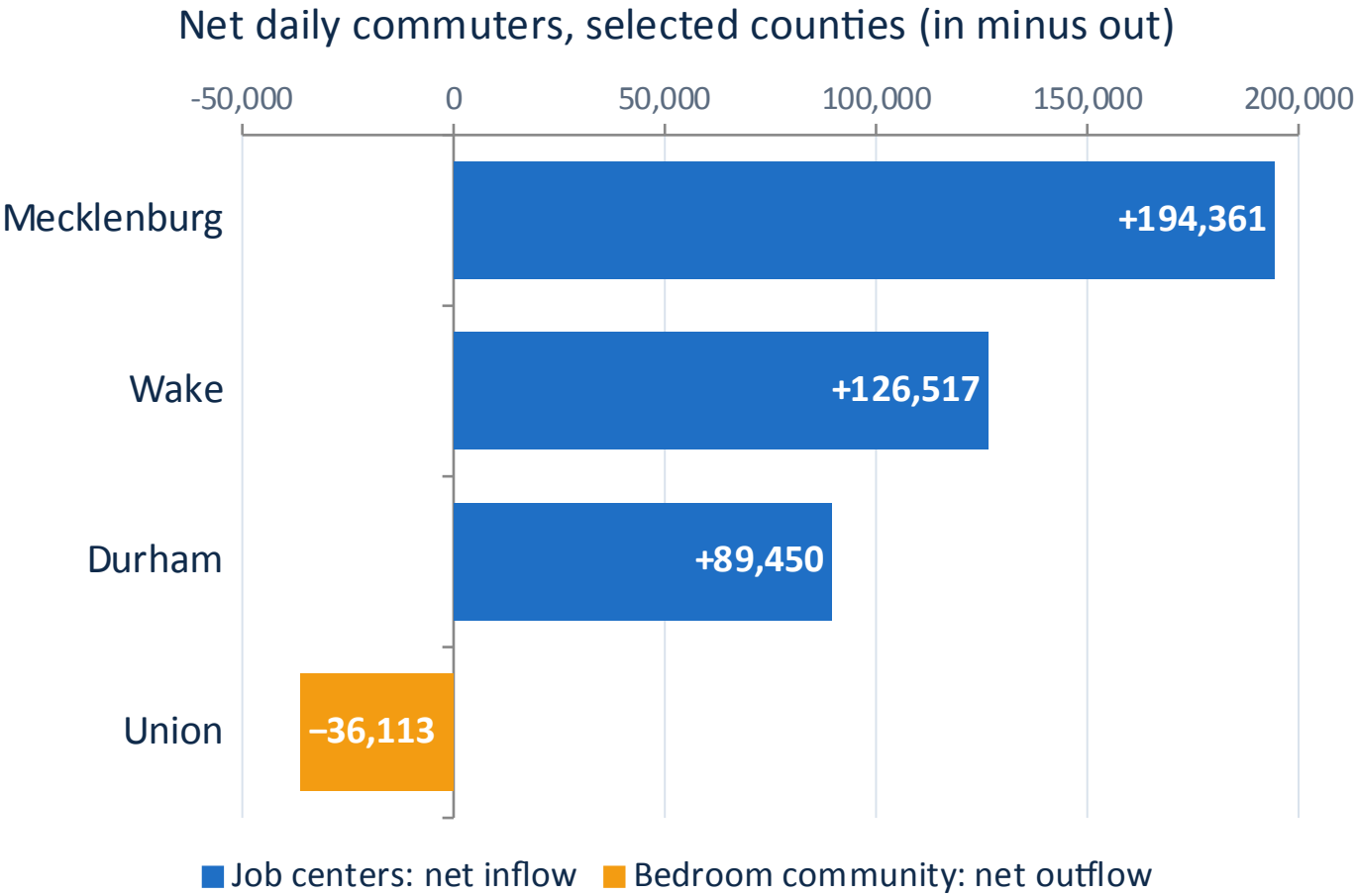
between

Exact additive decomposition (generalized entropy index $GE(1)$); verified to 2.22×10^{-16} ; Theil-L check: **82.5%** within

ZIP-level charging Gini (statewide, population-weighted): **0.566**

County-only formulas miss most of the problem → **two-tier design: county ranking + ZIP-level targeting**

Commuters Shift Demand Toward Employment Centers



U.S. Census commuting data (Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics, LEHD LODES, 2021): 4,198,163 workers → **859,260 adjusted commuters** after income and remote-work filters

Union sends more commuters out than it takes in: a bedroom community of Charlotte

Feeds the **cost-effectiveness pillar**: Mecklenburg scores highest (**0.801**)

Roughly Proportional Overall, with Strong County-by-County Variation

24.5%

of stations in the top 10 counties sit in disadvantaged tracts (296 of 1,210)

18.5%

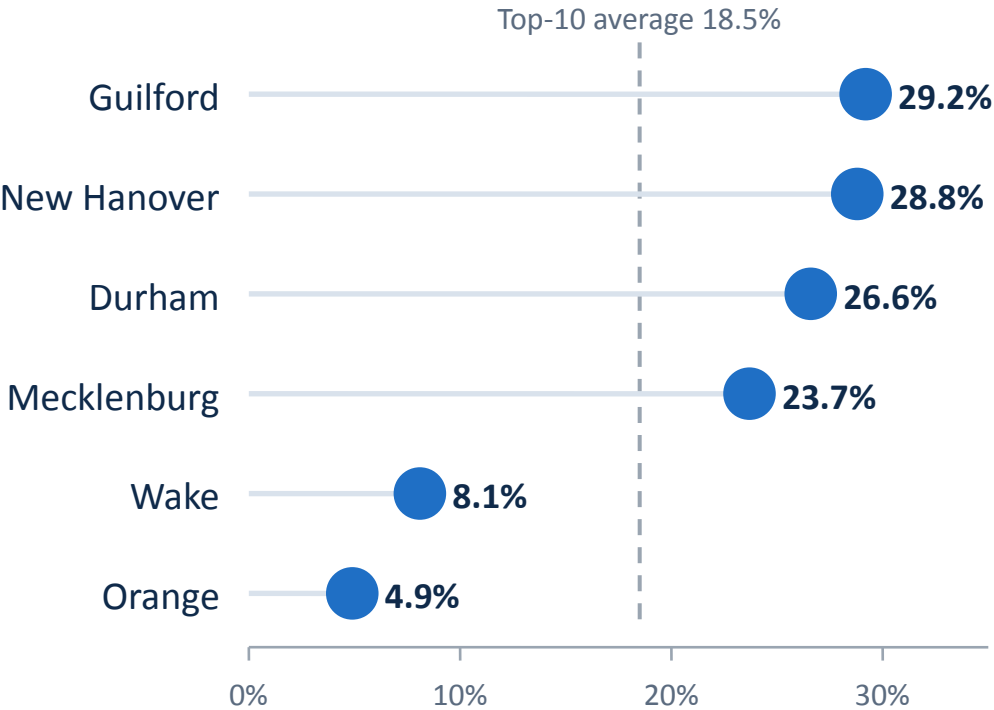
of residents in the top 10 counties live in those tracts

Roughly proportional overall

Statewide: **43.0%** of NC census tracts designated disadvantaged (934 of 2,170; CEJST v2.0)

Removing CEJST's climate category: **7 of 10** study counties unchanged

Residents in disadvantaged tracts, by county



CEJST v2.0 as of Jan 21, 2025; tool removed Jan 22, 2025 after Executive Order (EO) 14008 was rescinded; data from the Environmental Data & Governance Initiative (EDGI) / Public Environmental Data Partners (PEDP) archive

Three Pillars, One Score

$$\text{NEVI Priority Score} = 0.40 \times \text{Equity} + 0.35 \times \text{Utilization} + 0.25 \times \text{Cost-Effectiveness}$$

0.40 Equity

Disadvantaged residents, within-county charging Gini, underserved ZIPs; anchored on the Justice40 40% target (EO 14008, rescinded Jan 2025)

0.35 Utilization

BEVs per port; demand data validated out of sample (strongest data)

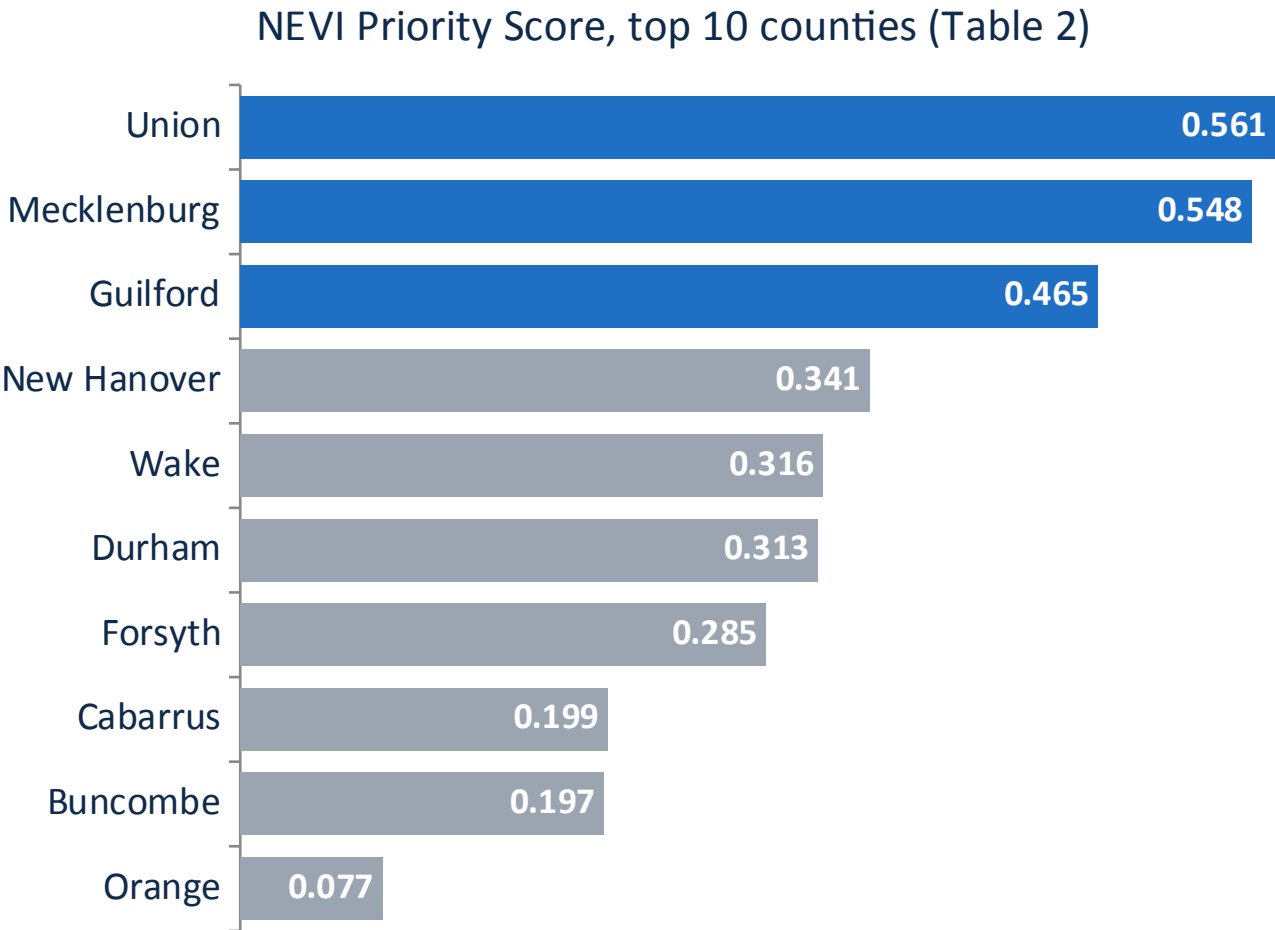
0.25 Cost-Effectiveness

Workplace demand (most uncertain input)

Weights follow data confidence. Each pillar is a weighted composite of sub-metrics, min-max normalized to 0–1.

Pillars measure different things: max variance inflation factor (VIF) 1.41 (concern threshold 5.0)

Top 3: Union, Mecklenburg, Guilford



0.561

#1 Union

0.548

#2 Mecklenburg

0.465

#3 Guilford

Top-3 set held; order shifts:

Equity weight 0.30–0.50 (five scenarios):
Mecklenburg gains and Union falls as the equity weight rises

Remote-work multiplier 0.75 / 0.85 / 0.95: cancels in min-max normalization, so the rankings are identical by construction

Wake ranks 5th despite the most BEVs (equity 0.322; 8.1% of residents in disadvantaged tracts)

Three County Archetypes

Union: utilization-driven

101.5 BEVs per port, more than 3× the next county → more stations

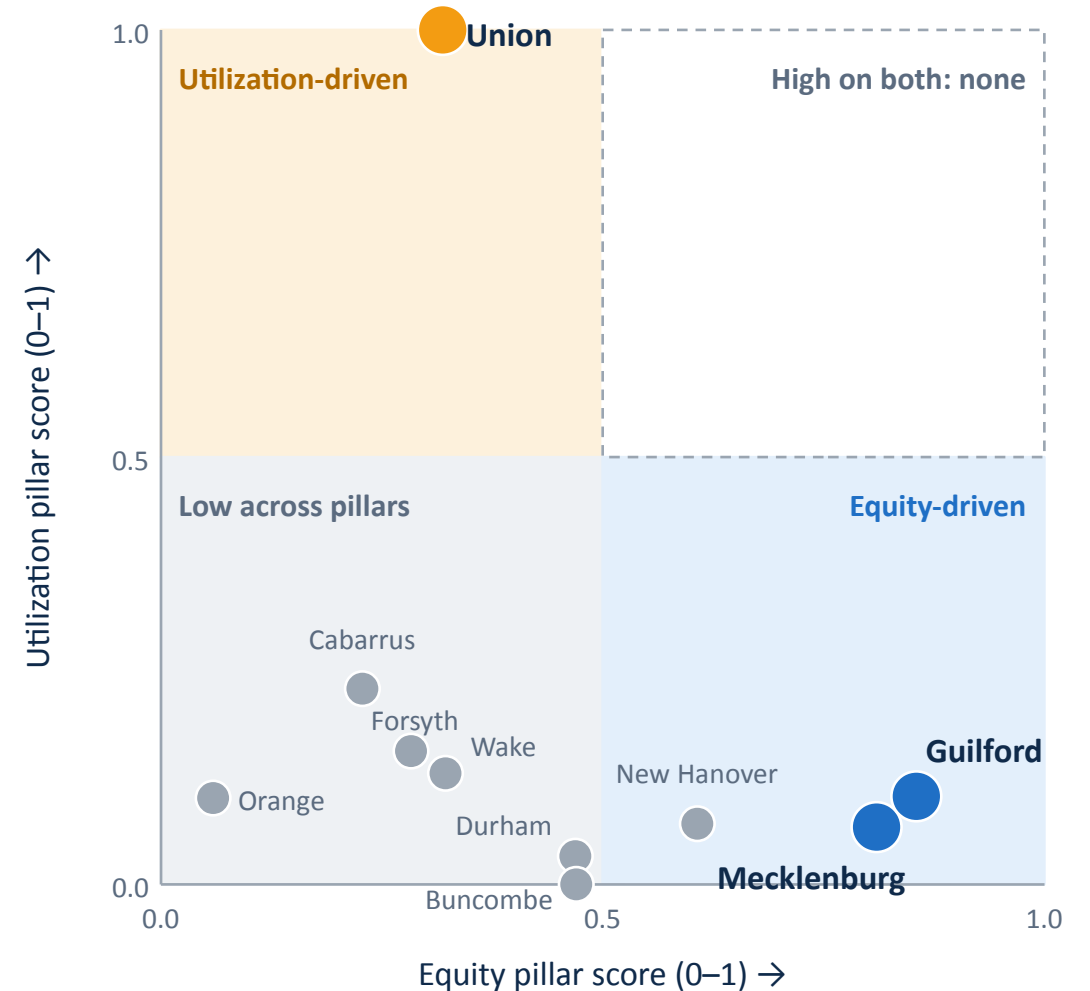
Mecklenburg & Guilford: equity-driven

Equity 0.810 and 0.855 → better-targeted stations in underserved ZIPs

Orange: low across all pillars

Lowest overall score (0.077) → no priority deployment now

Empty quadrant: no county is high on both equity and utilization



Contributions and Limitations

Contributions

1. Additive Theil-T decomposition of EV charging inequality, used as a design input to allocation
2. Area-weighted CEJST–AFDC crosswalk at ZIP resolution (12 checks, 23 of 23 sub-checks passed)
3. Three-pillar NEVI score with pillars that measure different things (VIF 1.41) and a robust top 3

Limitations, each with a direction of bias

Top-10 scope (73% of fleet): **bounded**

LODES 2021 vintage: **cancels in normalization**

Static BEV counts: **understates growth (conservative)**

CEJST v2.0 lock, NEVI rule changes: **direction unknown (policy risk)**

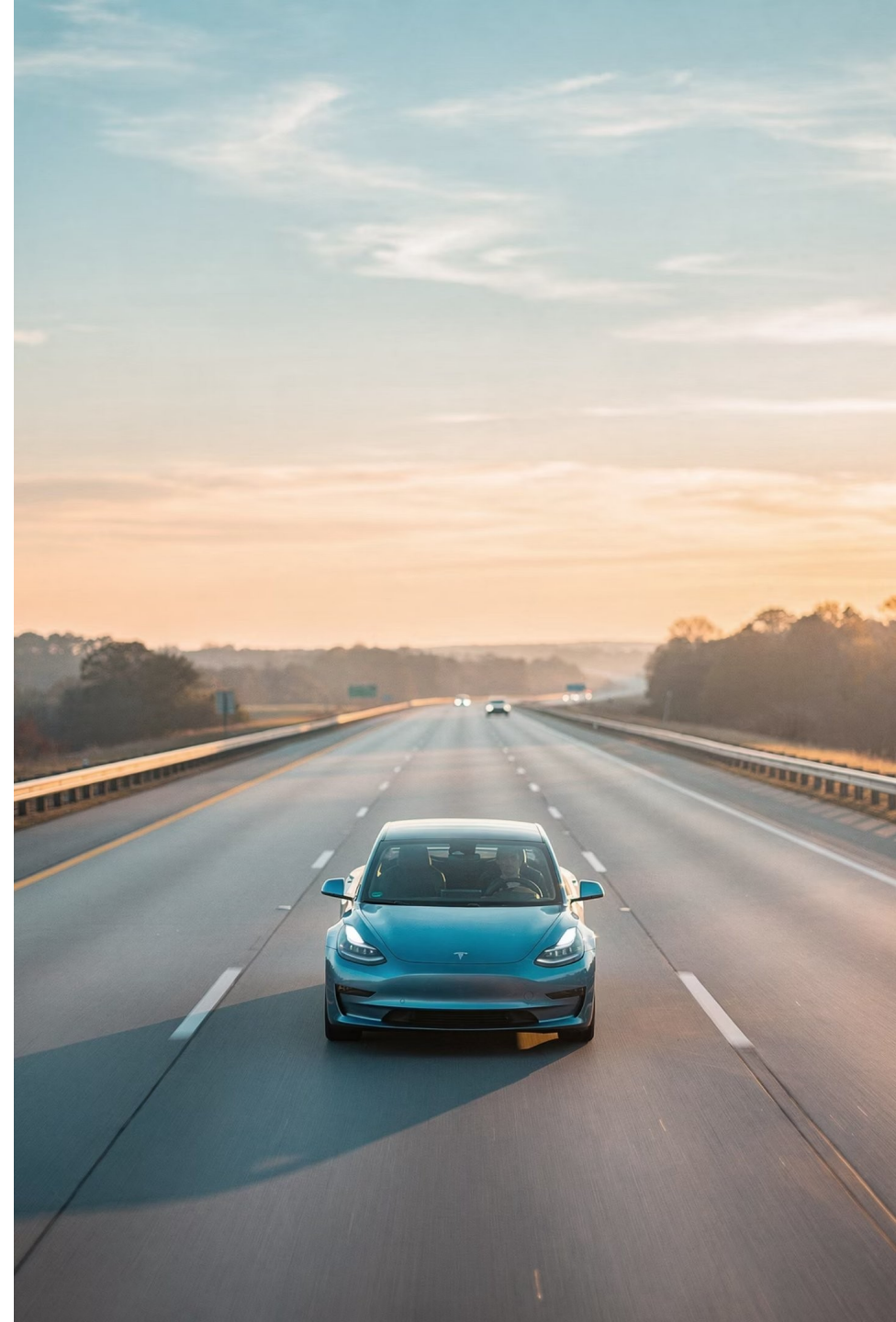
A Ranking to Inform NCDOT's Decisions, and What Comes Next

- ① **84.5% of charging inequality is within counties:** county rankings for where to invest, ZIP-level targeting for whom
- ② **Top 3:** Union 0.561 · Mecklenburg 0.548 · Guilford 0.465
- ③ **Forecasts hold out of sample (MAPE 4.34%) but run low:** plan an upward buffer of 4 to 5%

Next: all 100 counties · validate against NCDOT's actual NEVI deployments · confidence intervals on composite scores

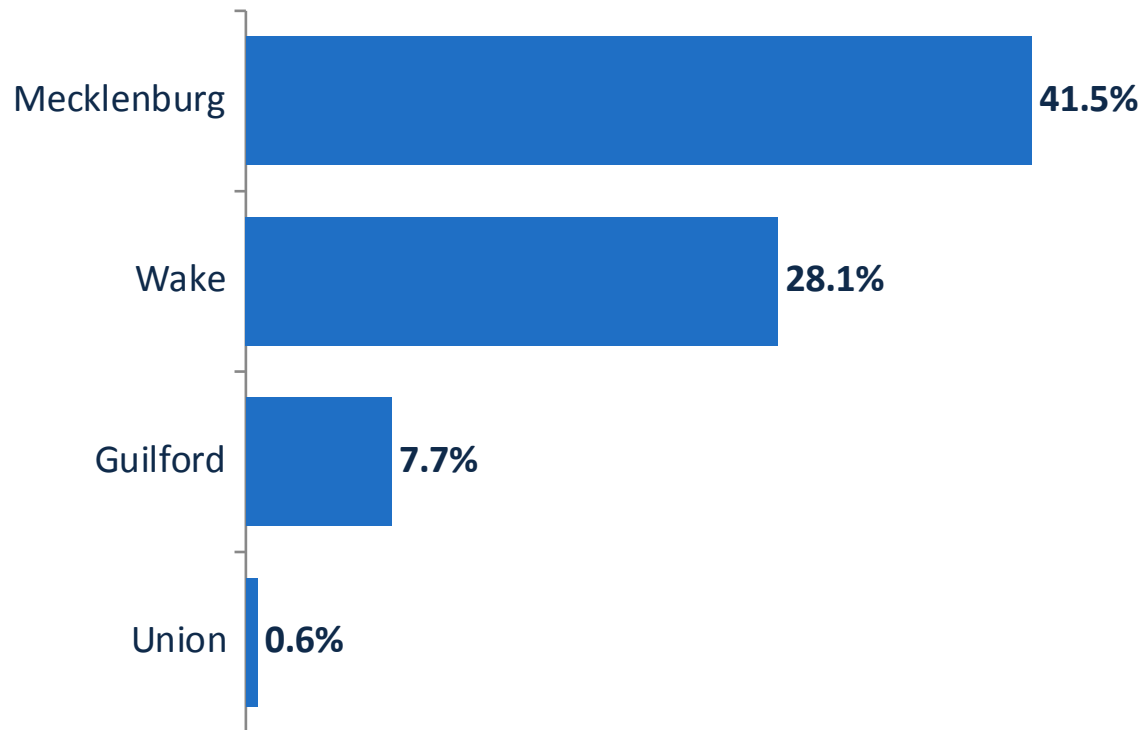
Thank you

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sanyer.org/research-lab



Theil-T Decomposition

Share of within-county inequality (the four counties reported in §6.3)



GE(1) / Theil-T is additively decomposable: **T = T_{between} + T_{within}**, with no residual (Bourguignon 1979; Shorrocks 1980)

Top 10 counties: **0.5791 = 0.0900 (15.5%) + 0.4892 (84.5%)**

Theil-L (GE(0)) robustness: **82.5%** within

Within-county contribution: Mecklenburg **41.5%**, Wake **28.1%**, Guilford **7.7%**, Union **0.6%**

Theil is a diagnostic for the architecture (why Tier 2 exists), not an input to either tier's score

Crosswalk Validation

CEJST v2.0 (2010 tracts) → 2020 ZCTAs, area-weighted in NC State Plane (EPSG:32119), the EPA EJScreen / HUD USPS approach

12 checks in 3 tiers, 23 of 23 sub-checks passed; per-ZCTA area conservation within 1%

Slivers < 100 m² dropped; border-state tracts (VA, SC, TN, GA) included; 25 zero-population tracts excluded

Known limit: uniform within-tract population (weaker in large rural tracts)

Full NEVI Scores (Table 2)

Rank	County	NEVI	Equity	Utilization	Cost-Eff.
1	Union	0.561	0.319	1.000	0.333
2	Mecklenburg	0.548	0.810	0.067	0.801
3	Guilford	0.465	0.855	0.103	0.347
4	New Hanover	0.341	0.607	0.071	0.293
5	Wake	0.316	0.322	0.130	0.568
6	Durham	0.313	0.469	0.033	0.454
7	Forsyth	0.285	0.283	0.156	0.467
8	Cabarrus	0.199	0.228	0.229	0.111
9	Buncombe	0.197	0.470	0.000	0.034
10	Orange	0.077	0.059	0.101	0.071

Equity weight 0.30–0.50 sweep: top 3 unchanged; Mecklenburg gains and Union falls as the equity weight rises; Orange stays 10th.

Data Sources and Diagnostics

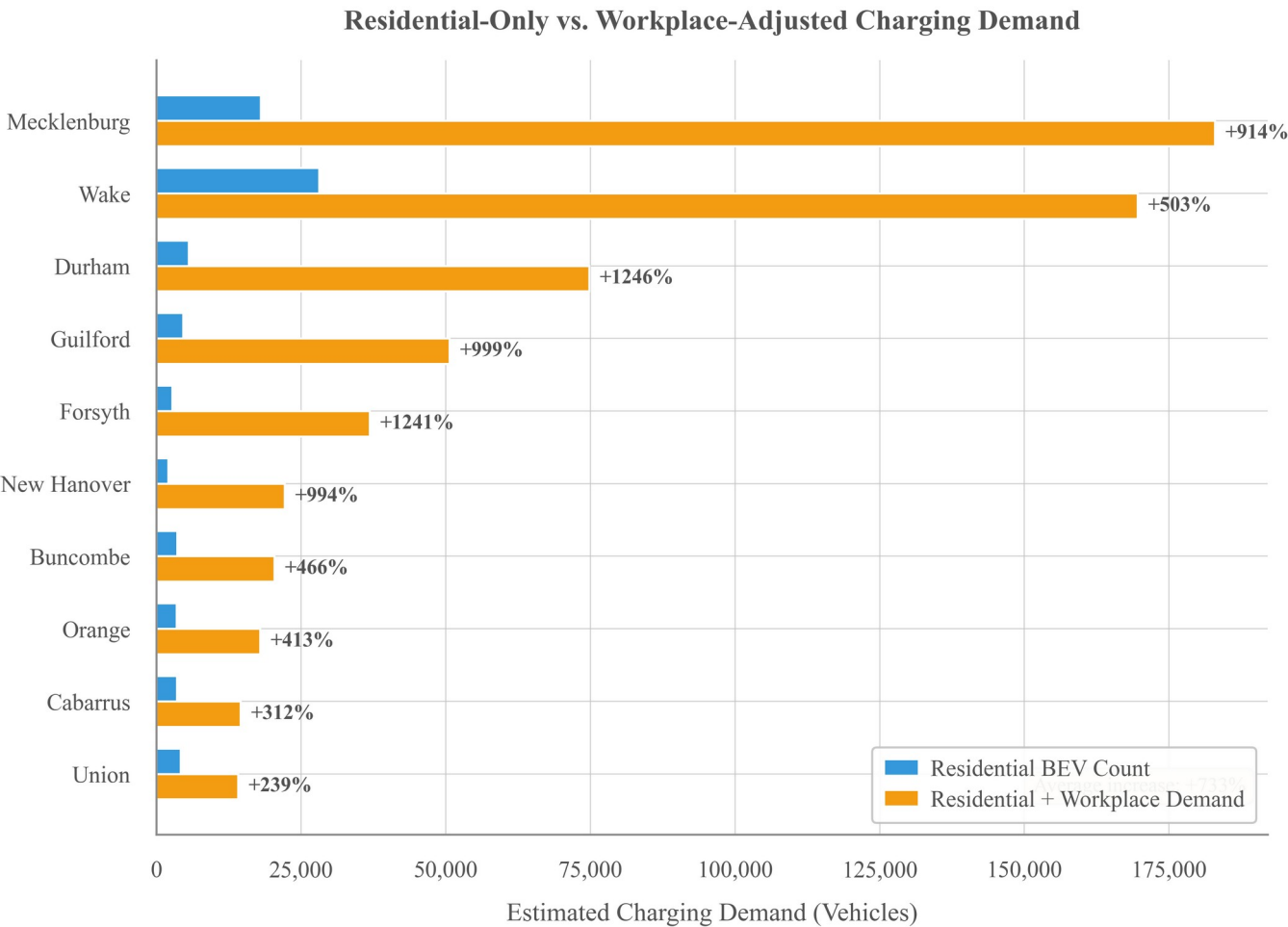
Source	Content	Window
NCDOT	Monthly county BEV registrations	Sep 2018 – Oct 2025
NREL AFDC API	Station inventory, all levels	Feb 2026 pull
U.S. Census ACS + TIGER	Income, tenure, population; boundaries	ACS 2022 five-year; TIGER 2020
LEHD LODES	Block-level commuter flows	2021
CEJST v2.0	Disadvantaged tract designations	Dec 2024 release (EDGI / PEDP archive)

CSS vs. MLE estimator: forecast difference < **0.1%**

NCDOT counting change (May 2025): ~**0.4%** of totals, works *against* the underprediction

Pillar correlations: E–U –0.26 · E–CE +0.48 · U–CE –0.03

Workplace Demand by County (Paper Fig 36)



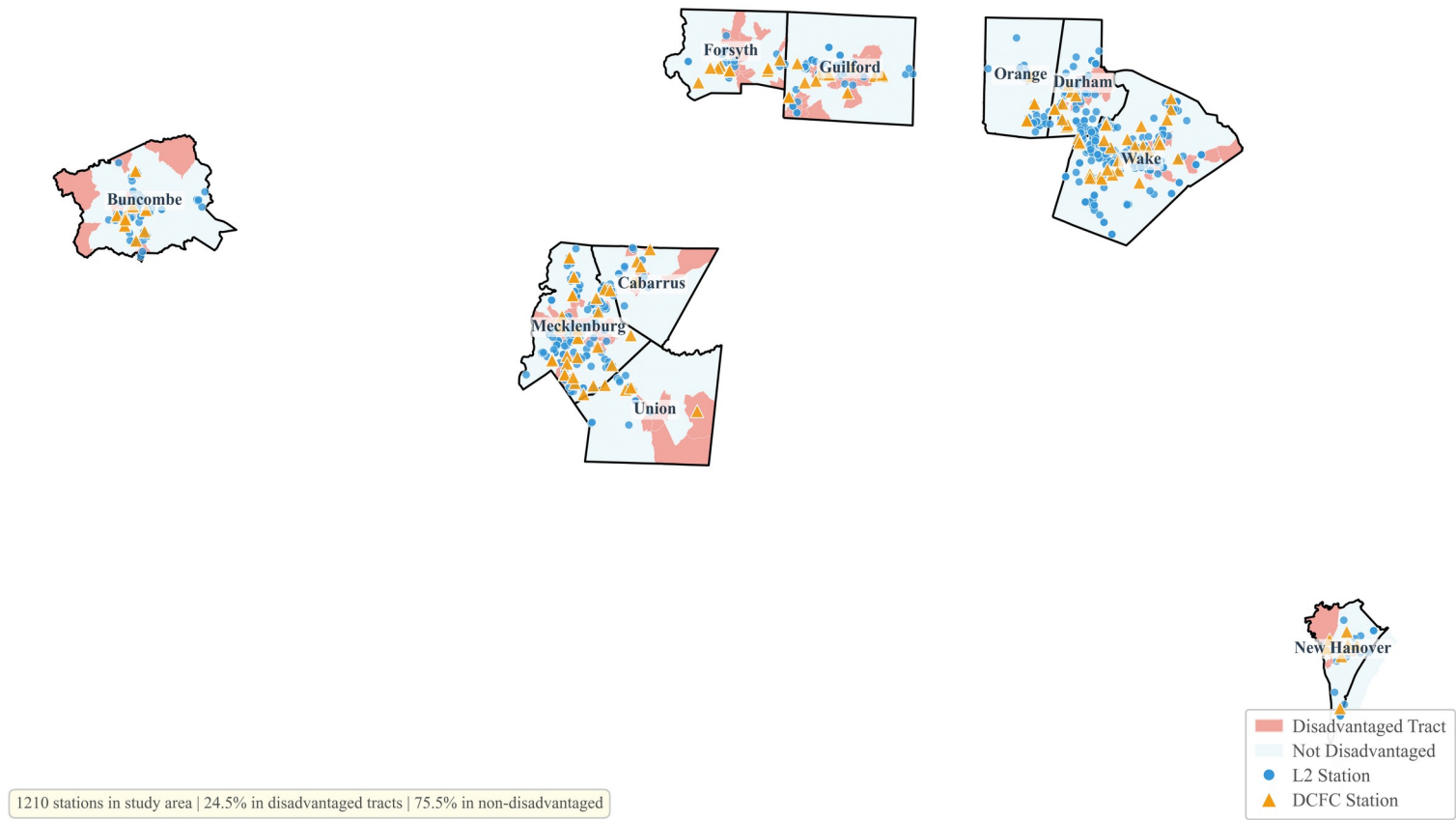
Blue: registered BEVs in the county (residential demand)

Orange: residential BEVs plus adjusted commuters who work in the county

The orange bar adds commuters to EVs: it shows **where daytime demand concentrates**, not an EV count. Port need uses a 30% charging adoption rate and a 15:1 port ratio (\$6.4).

Stations on Justice40 Tracts (Paper Fig 42)

EV Charging Stations on Justice40 Tracts — 10-County Study Area



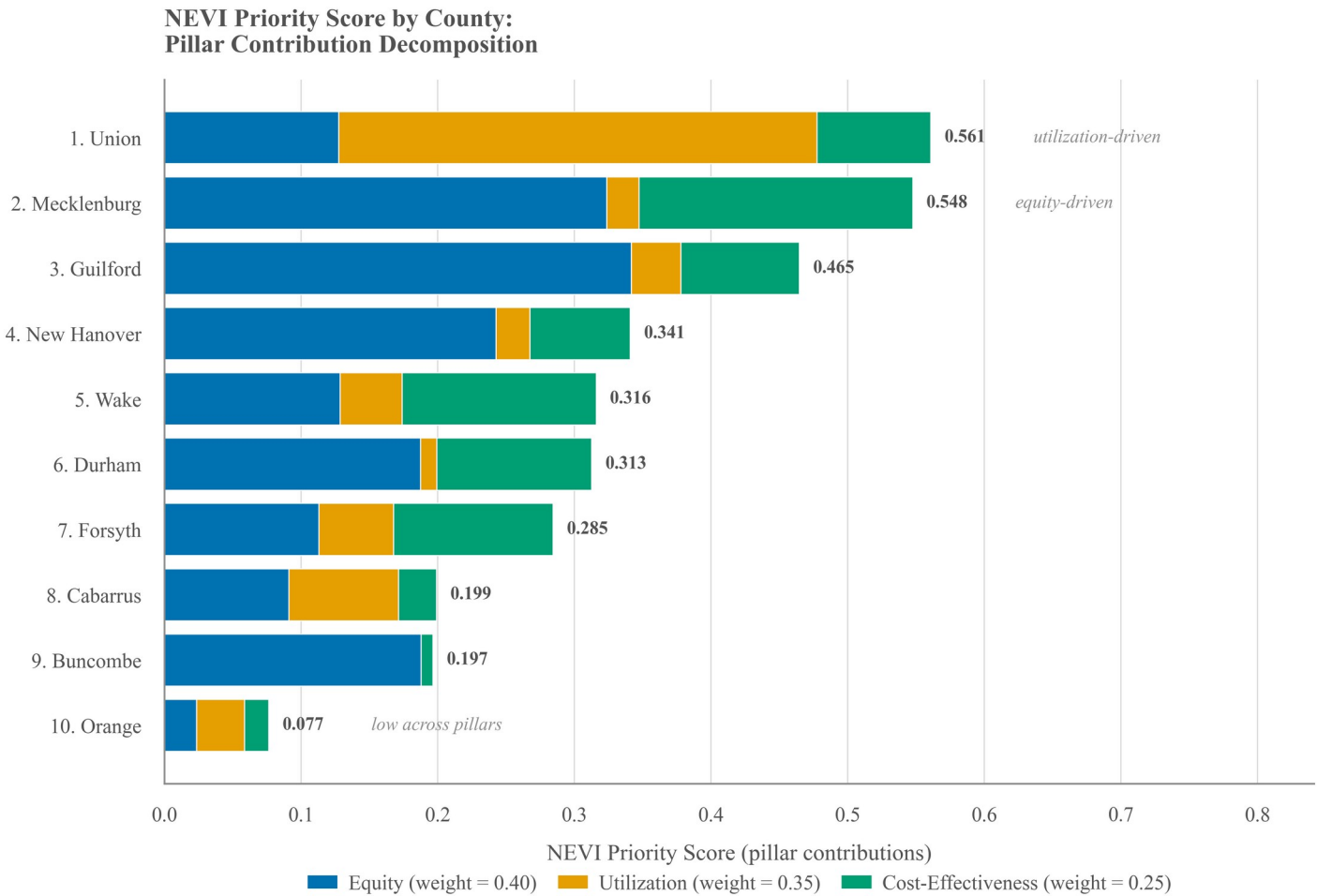
Pink: disadvantaged census tracts (CEJST v2.0)

Blue dots: Level 2 (L2) charging stations

Orange triangles: DC fast-charging (DCFC) stations

1,210 stations in the 10 study counties; 24.5% in disadvantaged tracts (\$6.5)

Pillar Contributions to the NEVI Score (Paper Fig 43)



Blue: equity (weight 0.40)

Orange: utilization (weight 0.35)

Green: cost-effectiveness (weight 0.25)

Union's bar is dominated by utilization;
Mecklenburg's and Guilford's by equity (\$6.7)

Tier 2: ZIP-Level Targeting

What Tier 2 uses: population, disadvantaged-community status, and station density, ranking ZIPs within each county (§4.4)

Its output: the top 20 underserved ZIPs hold **732,892 residents served by 58 ports**; **15** have zero DC fast chargers (§6.2)

Mecklenburg and Guilford each contribute **5 ZIPs** to the top 20: the state's two most infrastructure-rich counties also hold some of its worst gaps (§6.2)

Theil justifies Tier 2 but is **not an input** to either tier's score (§4.4)

Glossary of Abbreviations

Abbreviation	Meaning
ACS	American Community Survey
AFDC	Alternative Fuels Data Center
API	Application programming interface
ARIMA	Autoregressive integrated moving average
BEV	Battery electric vehicle
CEJST	Climate and Economic Justice Screening Tool
CSS / MLE	Conditional sum of squares / maximum likelihood estimation
DCFC	DC fast charging
E / U / CE	Equity / Utilization / Cost-Effectiveness pillars
EDGI	Environmental Data & Governance Initiative
EO	Executive Order
EPA EJScreen	U.S. Environmental Protection Agency environmental justice screening tool
EPSG	Standard ID code for a map projection (here, NC State Plane)
ESM	Exponential smoothing model
FHWA	Federal Highway Administration
GE(0) / GE(1)	Generalized entropy index, alpha = 0 (Theil-L) / alpha = 1 (Theil-T)

Abbreviation	Meaning
HUD USPS	Dept. of Housing and Urban Development / U.S. Postal Service ZIP crosswalk
IEA	International Energy Agency
IRA	Inflation Reduction Act
Justice40	Federal goal (EO 14008, 2021): 40% of certain federal benefits to disadvantaged communities; rescinded Jan 2025
L2	Level 2 charging
LEHD LODES	Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics
MAPE	Mean absolute percentage error
NCDOT	North Carolina Department of Transportation
NEVI	National Electric Vehicle Infrastructure (Formula Program)
NREL	National Renewable Energy Laboratory
PEDP	Public Environmental Data Partners
TIGER	Census Topologically Integrated Geographic Encoding and Referencing (boundaries)
UCM	Unobserved components model
VIF	Variance inflation factor
ZCTA	ZIP Code Tabulation Area